Object Detection

Holberton

What is Object Detection

Object Recognition

Object detection

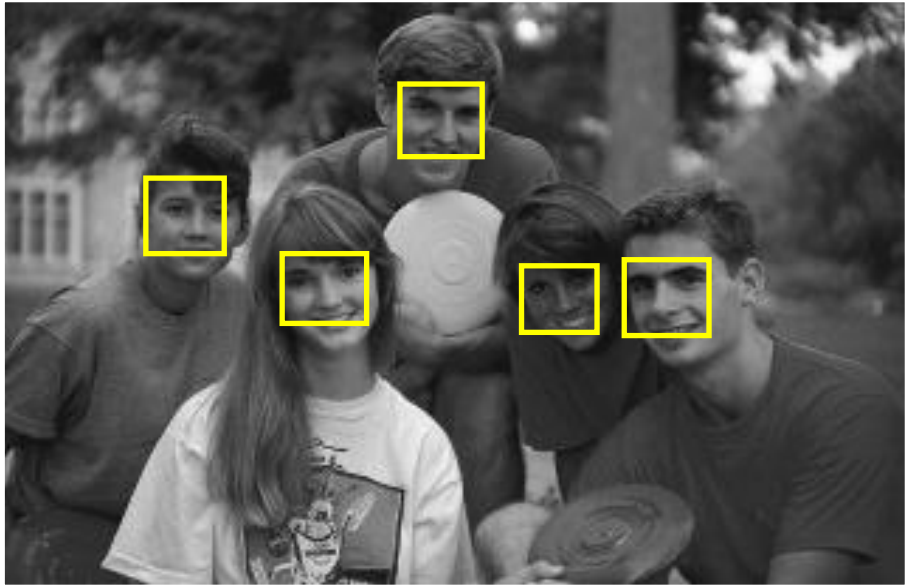
- Where is Jane?
- Where is a *face?*
- Is there a *face* in the image?

Object identification

- Who is it?
- Is it Jane or Erik?

Object Detection Problem Formulation

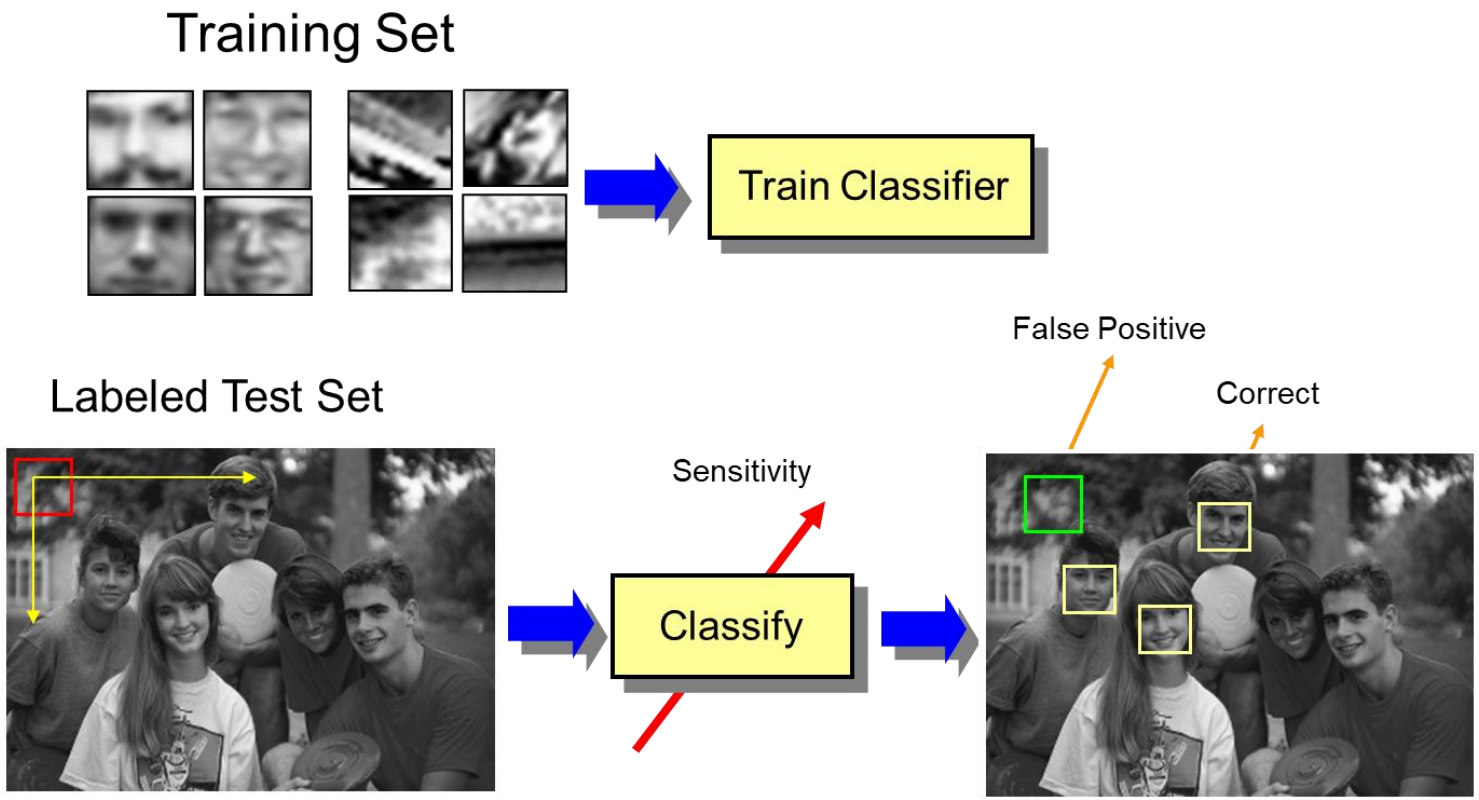
Given an input image, determine if there are objects of a given class (e.g., faces, people, cars) in the image and where are they located?



Object Detection

- Classifier must generalize over all exemplars of one class
- Negative class consists of everything else
- High accuracy required for most applications

Training and Testing in Object Detection



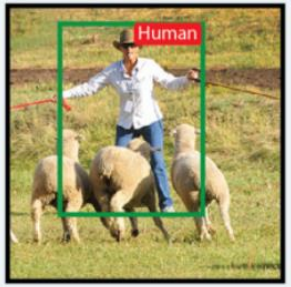
You Only Look Once (YOLO)

Common Tasks on Images

Image Classification

Classify an image based on the dominant object inside it.

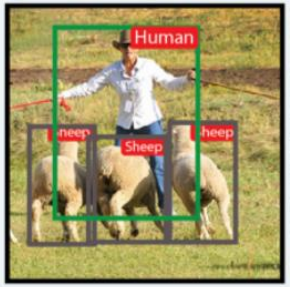
datasets: MNIST, CIFAR, ImageNet



Object Localization

Predict the image region that contains the dominant object. Then image classification can be used to recognize object in the region

datasets: ImageNet



Object Recognition

Localize and classify all objects appearing in the image. This task typically includes: proposing regions then classify the object inside them.

datasets: PASCAL, COCO

Semantic Segmentation

Label each pixel of an image by the object class that it belongs to, such as human, sheep, and grass in the example.

datasets: PASCAL, COCO

Instance Segmentation

Label each pixel of an image by the object class and object instance that it belongs to.

datasets: PASCAL, COCO

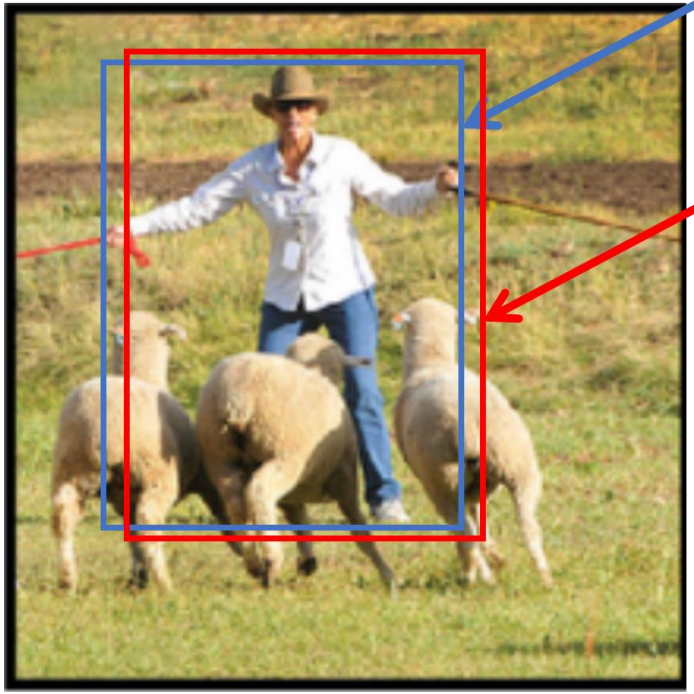
Keypoint Detection

Detect locations of a set of predefined keypoints of an object, such as keypoints in a human body, or a human face.

datasets: COCO

Bounding Box Proposal

Region of interest, region proposal, box proposal



Ground truth

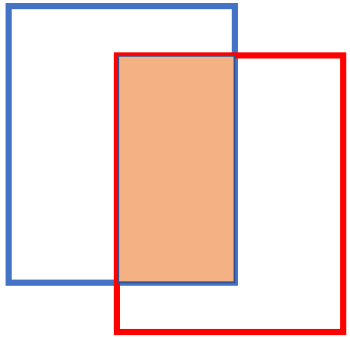
Proposed bounding box

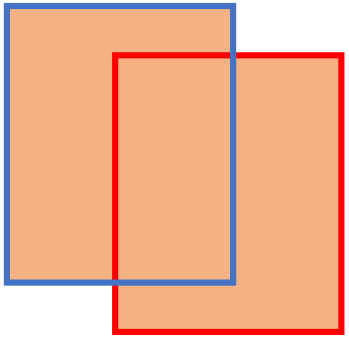
5 parameters

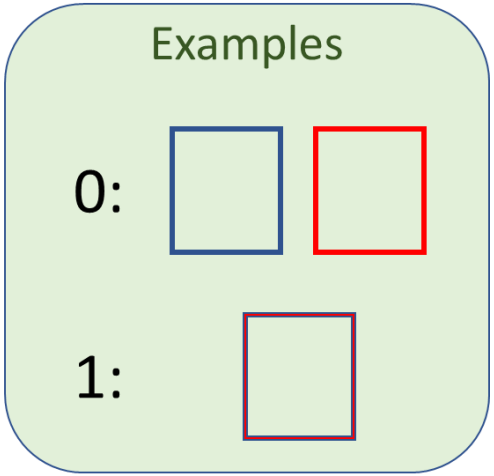
- w, h
- x, y
- confidence score: how likely it contains an object & accuracy of the box

How good: Intersection over Union (IOU)

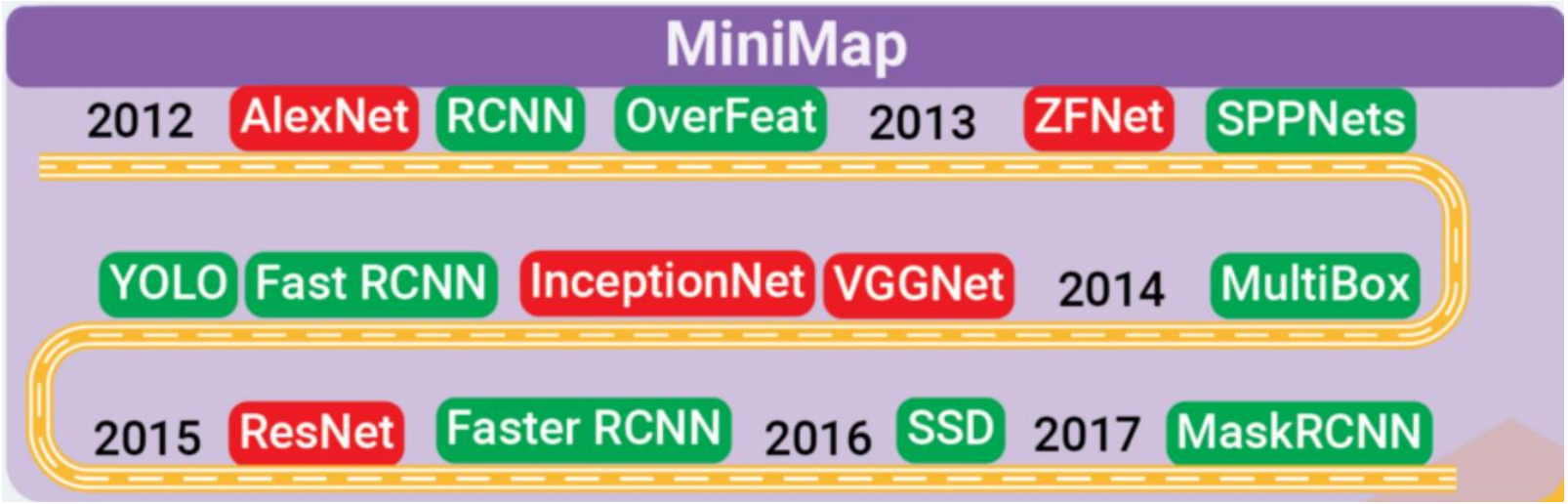
$$\text{IOU} = \frac{\text{Overlap Area}}{\text{Union Area}}$$







A brief history of object detection



Region proposal classification

- Accurate, but slow
- E.g. RCNN

Single shot:

- Region proposal + classification
- Fast, but less accurate
- E.g. YOLO, SSD

YOLO: You Only Look Once

Look once

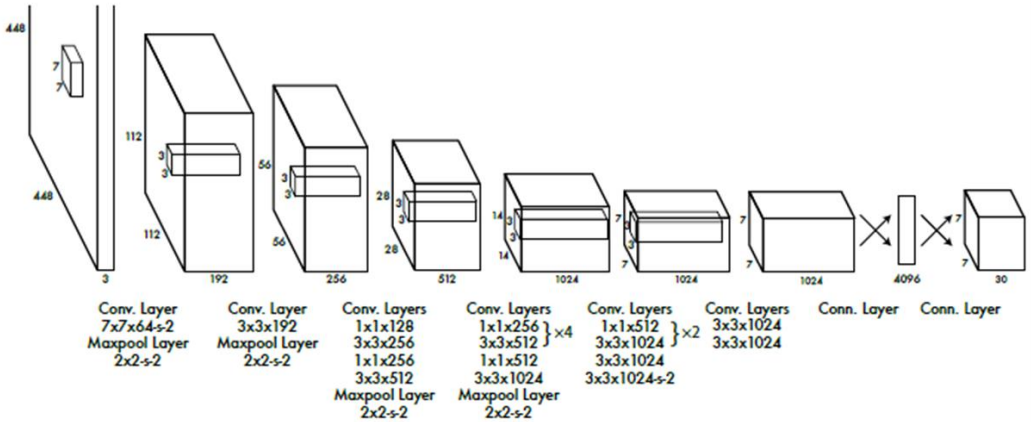
Results

- x, y, w, h
- confidence score:
  contain an object & box accuracy
- class score:
  belong to a class

Use CNN, since it's good enough

Why not regress? They are just numbers

YOLO: Let’s go CNN



YOLO v1's CNN: GoogLeNet variant, 24 layers

	Type	Filters	Size	Output
	Convolutional	32	3 × 3	256 × 256
	Convolutional	64	3 × 3 / 2	128 × 128
1×	Convolutional	32	1 × 1	
	Convolutional	64	3 × 3	
	Residual			128 × 128
	Convolutional	128	3 × 3 / 2	64 × 64
2×	Convolutional	64	1 × 1	
	Convolutional	128	3 × 3	
	Residual			64 × 64
	Convolutional	256	3 × 3 / 2	32 × 32
8×	Convolutional	128	1 × 1	
	Convolutional	256	3 × 3	
	Residual			32 × 32
	Convolutional	512	3 × 3 / 2	16 × 16
8×	Convolutional	256	1 × 1	
	Convolutional	512	3 × 3	
	Residual			16 × 16
	Convolutional	1024	3 × 3 / 2	8 × 8
4×	Convolutional	512	1 × 1	
	Convolutional	1024	3 × 3	
	Residual			8 × 8
	Avgpool		Global	
	Connected		1000	
	Softmax			

YOLO v3's CNN: darknet-53

Type	Filters	Size/Stride	Output
Convolutional	32	3 × 3	224 × 224
Maxpool		2 × 2/2	112 × 112
Convolutional	64	3 × 3	112 × 112
Maxpool		2 × 2/2	56 × 56
Convolutional	128	3 × 3	56 × 56
Convolutional	64	1 × 1	56 × 56
Convolutional	128	3 × 3	56 × 56
Maxpool		2 × 2/2	28 × 28
Convolutional	256	3 × 3	28 × 28
Convolutional	128	1 × 1	28 × 28
Convolutional	256	3 × 3	28 × 28
Maxpool		2 × 2/2	14 × 14
Convolutional	512	3 × 3	14 × 14
Convolutional	256	1 × 1	14 × 14
Convolutional	512	3 × 3	14 × 14
Convolutional	256	1 × 1	14 × 14
Convolutional	512	3 × 3	14 × 14
Maxpool		2 × 2/2	7 × 7
Convolutional	1024	3 × 3	7 × 7
Convolutional	512	1 × 1	7 × 7
Convolutional	1024	3 × 3	7 × 7
Convolutional	512	1 × 1	7 × 7
Convolutional	1024	3 × 3	7 × 7
Convolutional	1000	1 × 1	7 × 7
Avgpool		Global	1000
Softmax			

YOLO v2's CNN: darknet-19, 19 layers

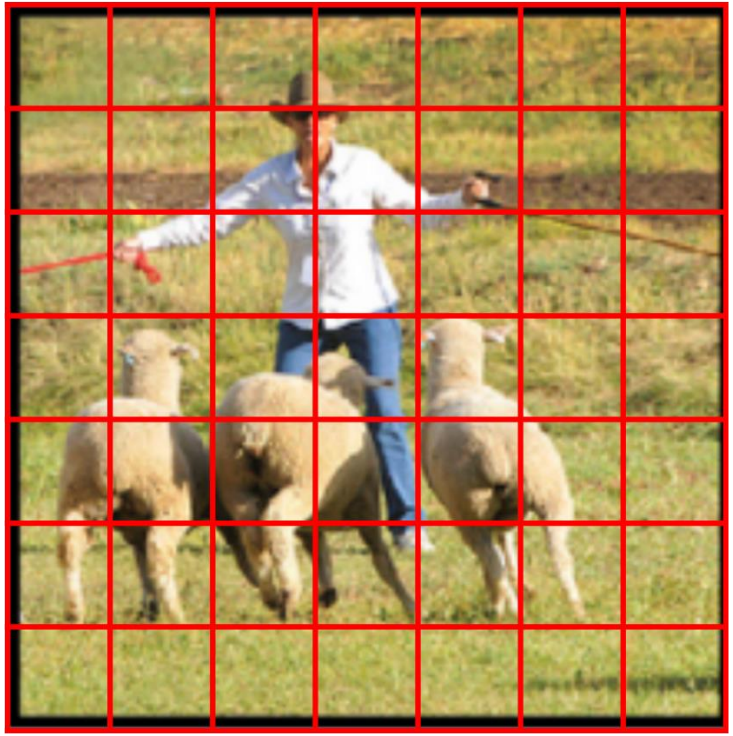
YOLO: Let’s do regression

Results for one box

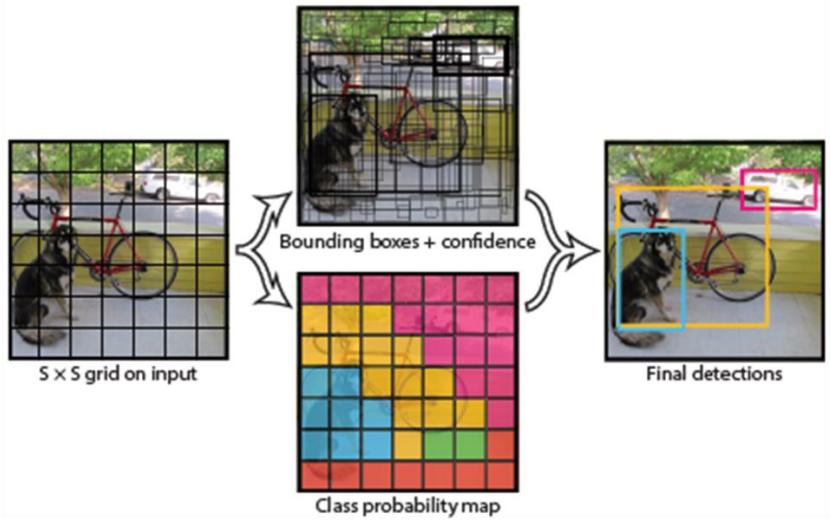
- x, y, w, h
- **Confidence score**: contain an object & box accuracy
- **Class score**: belong to a class

- **How to set up N?** (a large number? A random number?)

Better solution: use grids



YOLO: Let's do regression with non-maximal suppression



Grid 1	Proposed box 1 x, y, w, h confidence score	Proposed box 2 x, y, w, h confidence score	Class scores class 1 class 2, ... class 20
⋮	⋮	⋮	⋮
Grid SxS	Proposed box 1 x, y, w, h confidence score	Proposed box 2 x, y, w, h confidence score	Class scores class 1 class 2, ... class 20

We can use CNN to extract features, and finally perform a regression to detect objects.

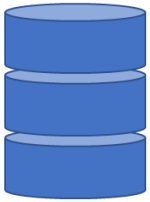
- YOLO v1: fully connected layers
- v2 & v3: convolutional layers

arXiv: 1506.02640, 1612.08242, 1804.02767

vector size: SxSx(5x2+20)

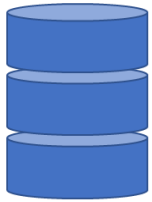
YOLO: Training

ImageNet:
classification dataset



YOLO

COCO/PASCAL VOC:
detection dataset



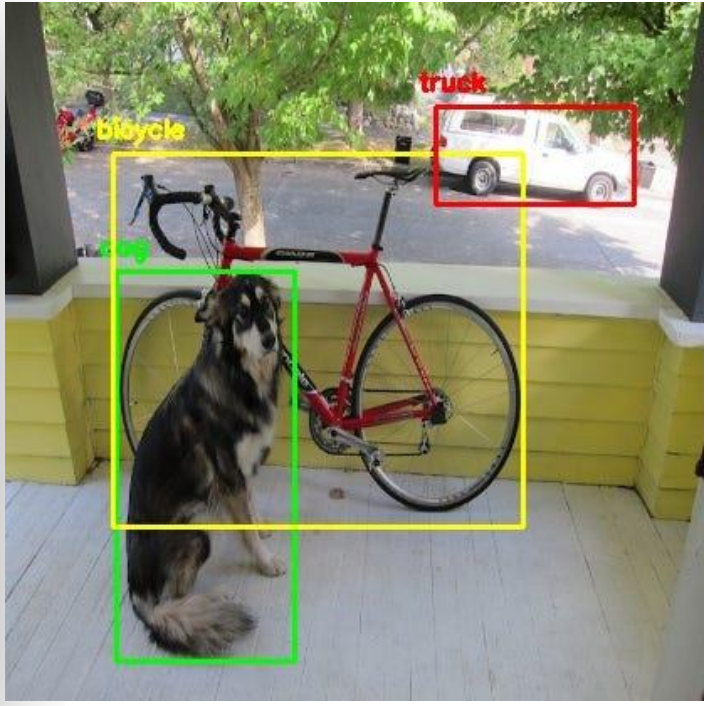
Step 1:
- train classification backbone

Step 2 (transfer learning):
- remove head layers
- add regression as new head
- fine-tune backbone & train head

Training tricks
- decaying learning rate
- batch normalization
- data augmentation

YOLO Algorithm

```
1. Divide image into a grid of cells

2. For each cell:
   - Predict multiple bounding boxes and their confidence scores
   - Apply a threshold to remove low-confidence boxes
   - Calculate final scores for remaining boxes

3. Select the box with the highest final score

4. Perform non-maximum suppression to eliminate overlapping boxes

5. Repeat for each cell in the grid
    - Upscale grid predictions to the original image size
    - Combine predictions from multiple scales
    - Apply a final confidence threshold
    - Optionally refine bounding boxes and estimate class probabilities

6. Output the final list of bounding boxes, labels, and scores
```



Any questions?